

Supplementary Information

Xibei Zhang^a, Shunzhou Wan^a, Agastya P. Bhati^a, Philip W. Fowler^{b,c}, Peter V. Coveney^{a,d,e*}

a. Centre for Computational Science, Department of Chemistry, University College London, London WC1H 0AJ, U. K

b. Nuffield Department of Medicine, John Radcliffe Hospital, University of Oxford, Oxford, UK

c. National Institute of Health Research Oxford Biomedical Research Centre, John Radcliffe Hospital, Oxford, UK

d. Advanced Research Computing Centre, University College London, London WC1H 0AJ, U.K.

e. Institute for Informatics, Faculty of Science, University of Amsterdam, 1098XH Amsterdam, The Netherlands

Correspondence

Peter V. Coveney

Email: p.v.coveney@ucl.ac.uk

The supplemental material provides additional analyses and data supporting the findings reported in the main text. Supplementary Table 1 presents the calculated effects of 61 clinically observed mutations in *Mycobacterium tuberculosis* RNA polymerase on rifampicin binding free energy, including calculated $\Delta\Delta G$ values, statistical errors, resistance/susceptibility classifications, and corresponding clinical phenotype data. Section S2 evaluates the proximity of missing loops in the reference crystal structure to the rifampicin binding site, demonstrating that reconstructed regions are distant from the binding pocket and unlikely to influence drug–protein interactions. Sections S3–S5 present further methodological evaluations, including assessments of sampling efficiency and λ -window mixing, quantitative convergence and flexibility analyses (RMSD & RMSF), and time-dependent convergence of $\Delta\Delta G$ for representative mutations. Together, these supplementary data provide methodological transparency, reproducibility, and additional context for interpreting the main results.

S1. Calculated impact of the 61 mutations on rifampicin's binding free energy to RNAP

This part tabulates mutations, their binding free energy changes ($\Delta\Delta G \pm \text{error}$), resistance/susceptibility classifications (R/S), and clinical resistance data. Key mutations include phenotype ratio and sample sizes.

Supplementary Table 1 The calculated impact of the 61 mutations on rifampicin's binding free energy to RNAP is presented. Exp. represents clinical resistance results, while $\Delta\Delta G$, Error, R/S reflect calculations from our TIES_PM study. All values for calculations are in kcal/mol.

NO.	Mut.	$\Delta\Delta G$	Err.	R/S	Exp.	Phenotype Ratio	Sample Size
1	D435F	8.6	2.5	R	R	90.7%	43
2	D435V	4.9	2.3	R	R	98.8%	1096
3	H445C	-0.1	1.9	U	R	91.5%	71
4	H445D	9.3	3.8	R	R	98.4%	574
5	H445L	5.6	1.7	R	R	91.4%	210
6	H445R	8.3	3.2	R	R	98.1%	107
7	H445Y	6.4	2.3	R	R	98.6%	484
8	I491F	-1.2	1.6	S	R	55.5%	173
9	L452P	-1.6	1.5	S	R	64.9%	259
10	Q432K	17.7	2.5	R	R	100.0%	21
11	Q432L	4.9	1.1	R	R	94.1%	17
12	Q432P	5.8	2.1	R	R	98.1%	52
13	S441L	4.4	3.0	R	R	94.4%	18

14	S441Q	2.9	3.3	U	R	93.8%	32
15	S450F	6.2	1.5	R	R	100.0%	195
16	S450L	5.6	1.3	R	R	97.9%	8912
17	S450Q	5.5	1.9	R	R	100.0%	9
18	S450W	4.8	2.0	R	R	97.4%	192
19	V170F	6.7	1.5	R	R	97.6%	41

33

NO.	Mut.	$\Delta\Delta G$	Err.	R/S	Exp.	Phenotype Ratio	Sample Size
20	E250G	0.2	1.2	S	S	8.4%	83
21	I925V	-0.1	0.7	S	S	1.4%	70
22	K944E	-0.7	0.7	S	S	-	0
23	N381H	0.0	1.6	S	S	100.0%	76
24	R511I	-1.7	2.8	S	S	-	-
25	S388L	0.3	1.0	S	S	100.0%	1
26	S428C	0.1	1.0	S	S	-	-
27	V109I	-0.8	0.7	S	S	-	-
28	V179A	-0.3	0.6	S	S	-	-
29	A451V	0.3	1.4	S	U	100.0%	3

30	D435A	5.4	2.1	R	U	100.0%	3
31	D435G	5.1	2.3	R	U	50.0%	8
32	G426S	-0.2	0.5	S	U	100.0%	1
33	H445F	4.7	2.0	R	U	100.0%	2
34	H445N	2.2	1.9	U	U	33.7%	104
35	H445Q	2.9	2.0	U	U	100.0%	2
36	K446E	1.2	3.5	U	U	100.0%	1
37	L430P	0.2	1.2	U	U	25.5%	208
38	L443F	0.7	1.3	U	U	100.0%	1
39	M434I	0.1	1.7	U	U	100.0%	1
40	R448K	5.3	5.6	U	U	100.0%	4
41	S431T	2.6	1.5	R	U	-	-

34

35

NO.	Mut.	$\Delta\Delta G$	Err.	R/S	Exp.	Phenotype Ratio	Sample Size
42	S450V	-0.2	1.1	S	U	100.0%	4
43	T427N	-0.9	0.9	S	U	100.0%	2
44	A334D	1.3	2.1	U	U	100.0%	1
45	D270E	1.1	2.6	U	U	100.0%	2
46	E132D	-0.4	2.3	U	U	100.0%	2
47	G28R	0.3	2.1	U	U	100.0%	6
48	G981D	-0.1	2.1	U	U	100.0%	1
49	H343Q	-1.2	2.0	S	U	100.0%	21
50	M390T	-1.4	1.4	S	U	5.3%	19
51	P30S	-0.1	1.2	S	U	100.0%	2
52	P454L	0.3	0.6	S	U	100.0%	6
53	S195R	1.0	1.9	U	U	100.0%	1
54	T350I	-0.6	1.0	S	U	100.0%	22
55	T526S	-1.2	2.3	S	U	100.0%	1
56	T585A	-0.3	1.4	S	U	-	-
57	T676P	0.3	1.5	U	U	-	-
58	V113I	-0.2	0.6	S	U	7.1%	53

59	V359A	-0.1	0.7	S	U	7.1%	14
60	V695L	-0.3	0.8	S	U	9.5%	74
61	V77M	-0.2	0.7	S	U	100.0%	1

36

37

38

39 **S2. Assessment of Missing Loop Proximity to the Rifampicin Binding Site**

40 To ensure that missing loops in the crystal structure (PDB ID: 5UH6) do not affect rifampicin
41 binding, we evaluated their spatial proximity to the ligand by measuring distances between
42 flanking resolved residues and rifampicin atoms. The process is described below to ensure
43 reproducibility:

44 **(1). Identifying Missing Loops**

45 Missing residues were identified by inspecting the REMARK 465 records in the PDB file.
46 Specifically:

47 - Chain B: Residues 155–157 are missing.

48 - Chain D: Residues 1012–1024 are missing.

49 These loops were later reconstructed using AlphaFold2-predicted models (UniProt IDs:
50 P9WGZ1 for chain B and P9WGY7 for chain D).

51 **(2). Reference Ligand Atom**

52 The rifampicin molecule is designated as residue 1201 on chain C with the residue name RFP.
53 Atom C1 within RFP was selected as a reference point for distance measurements.

54 **(3). Distance Measurement Procedure**

55 All measurements were performed using VMD 1.9.4 via the Tk Console, based on the
56 coordinates of the resolved residues and the ligand. The following commands were used:

```
57 # Select rifampicin reference atom
58 set sel_rfp [atomselect top "resname RFP and chain C and resid 1201 and name C1"]
59 set pos_rfp [lindex [$sel_rfp get {x y z}] 0]
60 # For chain B: flanking residue before missing loop
61 set sel_B154 [atomselect top "chain B and resid 154 and name CA"]
62 set pos_B154 [lindex [$sel_B154 get {x y z}] 0]
63 set dist_B [veclength [vecsub $pos_B154 $pos_rfp]]
64 puts "Distance from B:154 (CA) to RFP C1: $dist_B Å"
65 # For chain D: flanking residues around the missing loop
66 set sel_D1011 [atomselect top "chain D and resid 1011 and name CA"]
67 set pos_D1011 [lindex [$sel_D1011 get {x y z}] 0]
68 set dist_D_pre [veclength [vecsub $pos_D1011 $pos_rfp]]
```

```

69     puts "Distance from D:1011 (CA) to RFP C1: $dist_D_pre Å"
70     set sel_D1026 [atomselect top "chain D and resid 1026 and name CA"]
71     set pos_D1026 [lindex [$sel_D1026 get {x y z}] 0]
72     set dist_D_post [veclength [vecsub $pos_D1026 $pos_rfp]]
73     puts "Distance from D:1026 (CA) to RFP C1: $dist_D_post Å"

```

74 **(4). Results**

75 The measured distances were:

76 - Chain B: B:154 (CA) → RFP C1 = 82.75 Å

77 - Chain D: D:1011 (CA) → RFP C1 = 29.78 Å

78 D:1026 (CA) → RFP C1 = 35.37 Å

79 These values confirm that the reconstructed loop regions are spatially distant from the
80 binding pocket and are not expected to affect drug–protein interactions.

81

82

83

S3. Evaluations of Sampling Efficiency and λ -Window Mixing in Independent Alchemical Simulations

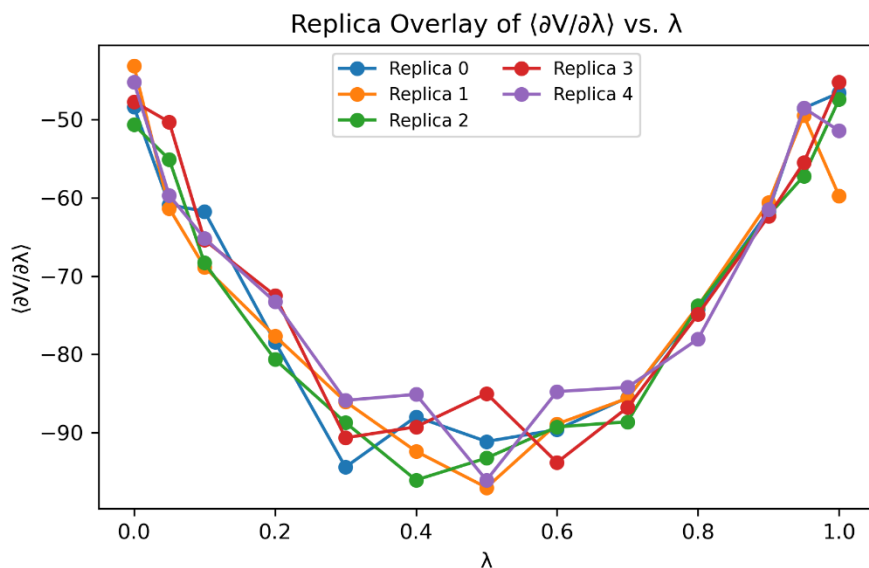


Figure S3. 1 Replica Overlay of $\langle \partial V / \partial \lambda \rangle$ vs. λ

Each colored line shows one of the five independent replicas. The smooth, U-shaped curves and the close agreement among replicas at all 13 λ points indicate that each window is well sampled and that there are no large outliers in any replica.

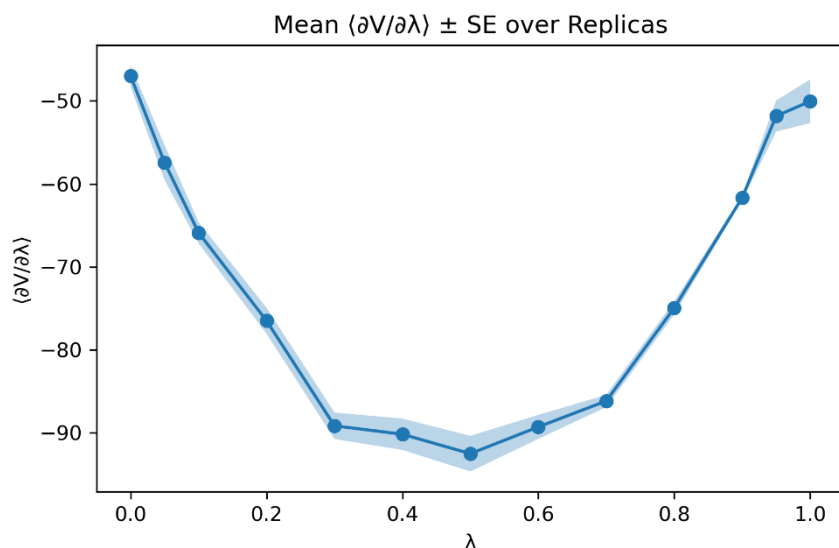
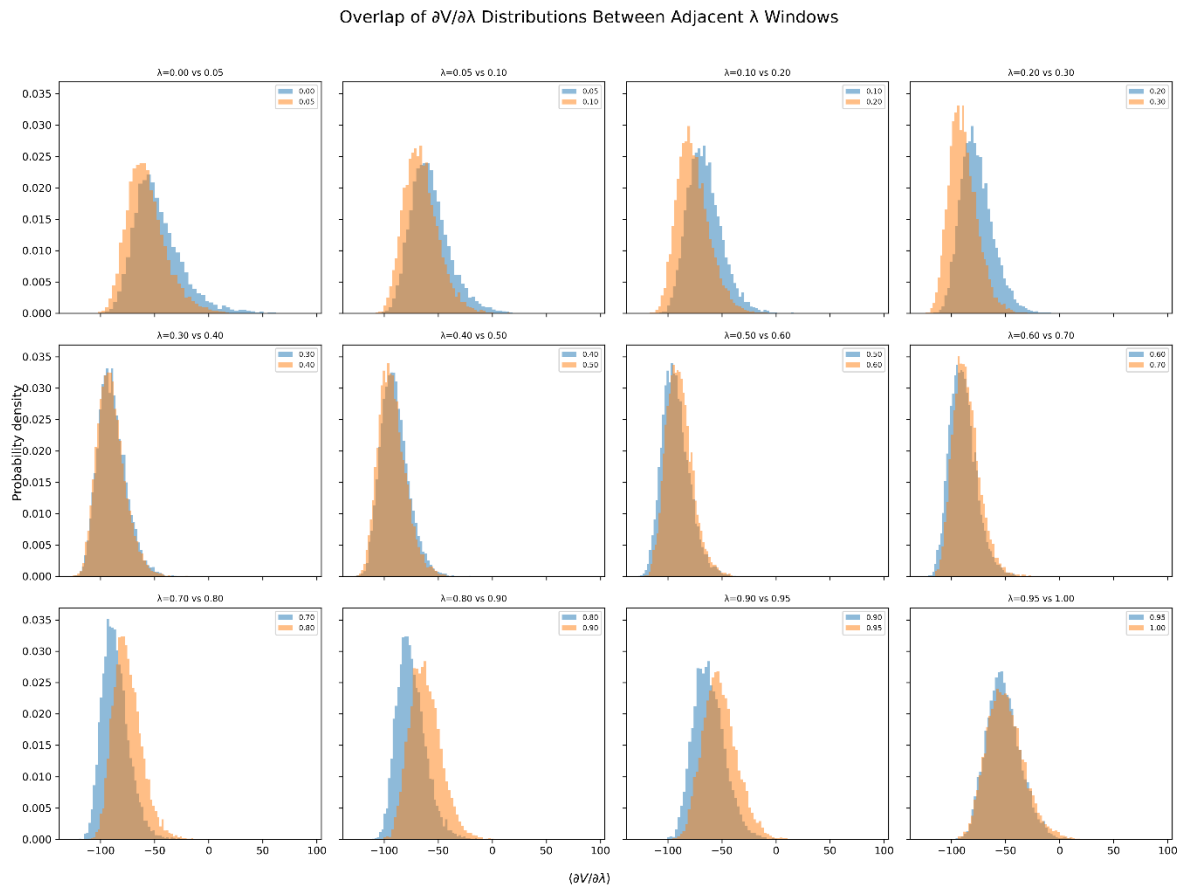


Figure S3. 2 Mean $\langle \partial V / \partial \lambda \rangle \pm \text{Standard Error}$ over Replicas

The dark line is the grand mean across all five replicas; the shaded band is $\pm \text{SE}$. The narrow error band demonstrates very low inter-replica variability and hence high precision in the

95 integral estimate.



96

97

Figure S3.3 Overlap of $\partial V/\partial \lambda$ Distributions Between Adjacent λ Windows

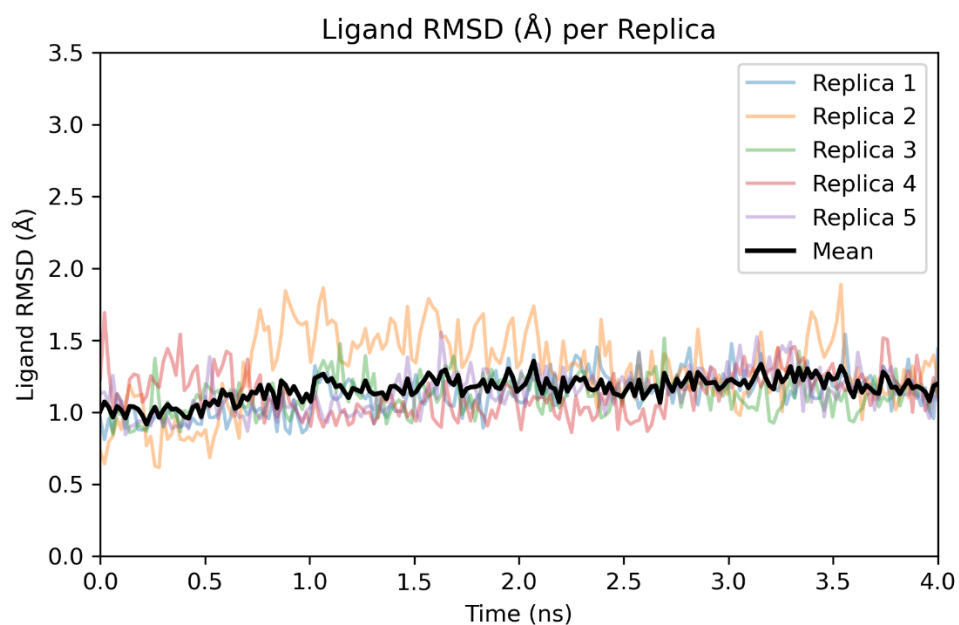
98

Each subplot shows semi-transparent histograms of the combined samples from all replicas at two neighboring λ values. Substantial overlap in every panel (typically $>20\%$) confirms that neighboring windows share phase-space coverage and mixing is adequate for reliable free-energy integration.

102

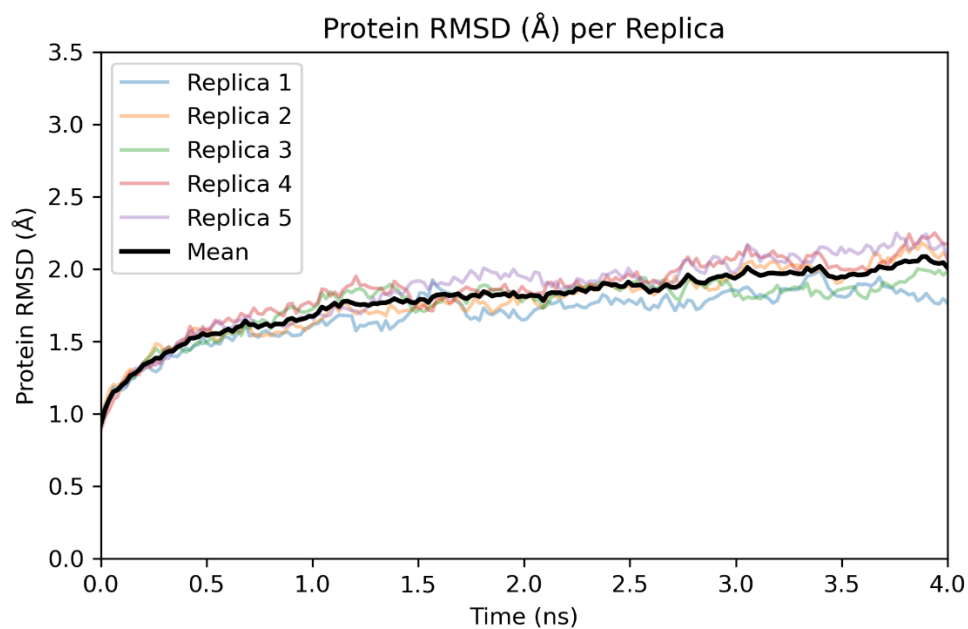
103

104 S4. Quantitative Convergence and Flexibility Analysis (RMSD & RMSF)



105

106 Figure S4. 1 Ligand RMSD across five independent replicas over 4 ns. It remains tightly clustered at 1.0–1.5 Å,
107 confirming stable binding.



108

109 Figure S4. 2 Protein backbone RMSD across five independent replicas over 4 ns. It equilibrates to 1.6–2.2 Å
110 within 4 ns in each replica, demonstrating rapid convergence.

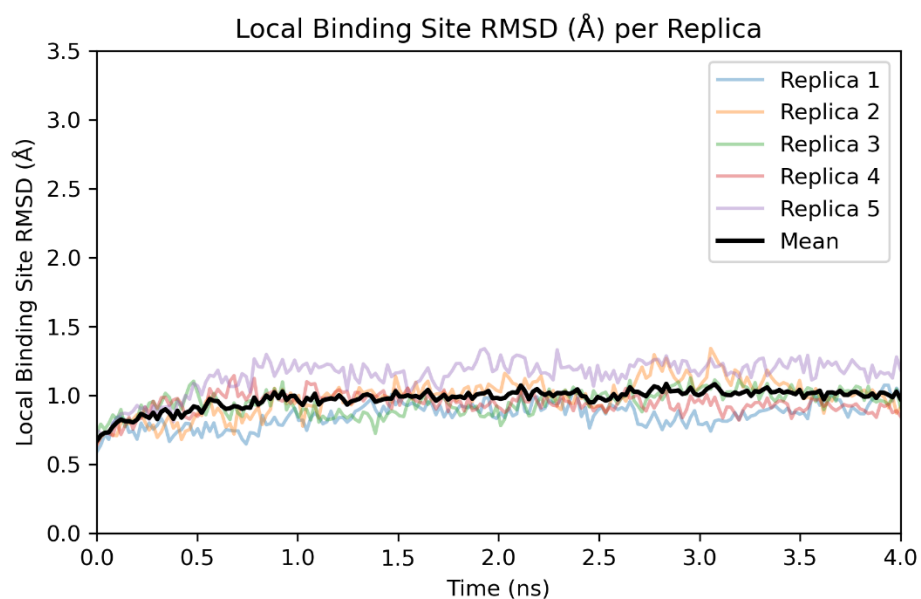


Figure S4. 3 Local protein backbone RMSD (within 10 Å of RFP) across five replicas over 4 ns. The RMSD converges around 1.0 Å, indicating stable local geometry near the binding site.

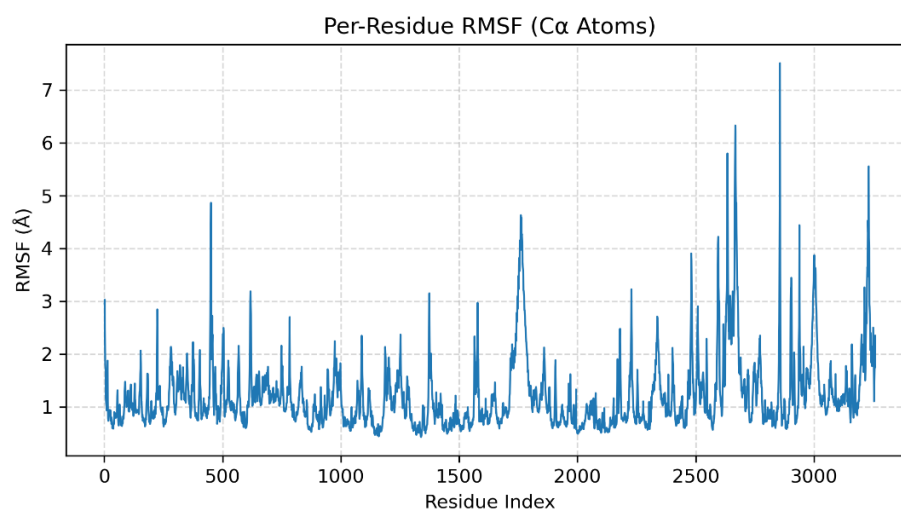
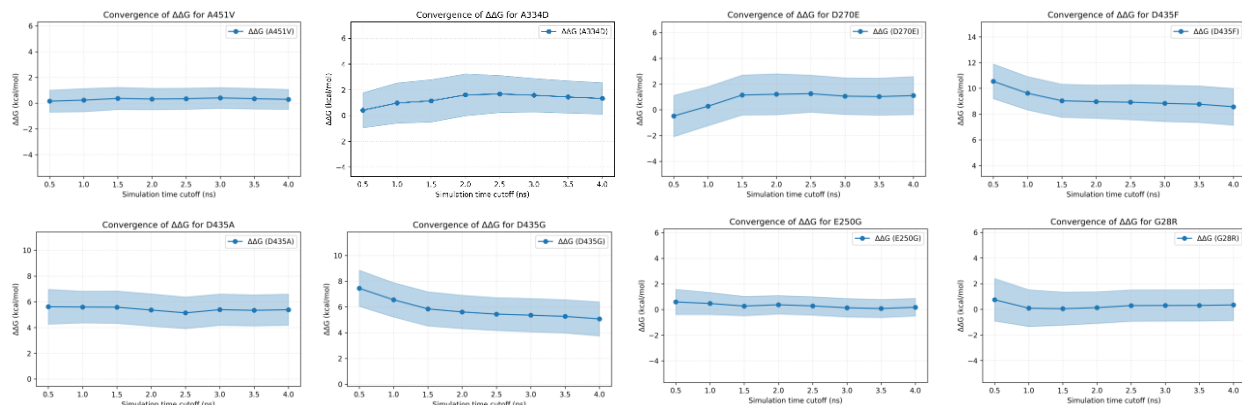


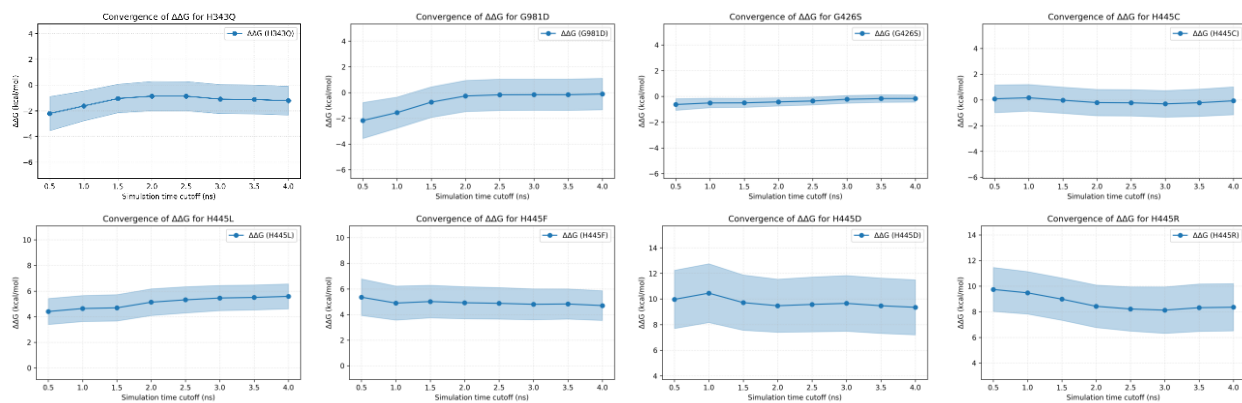
Figure S4. 4 Per-residue C α RMSF showing core rigidity and loop flexibility

120 S5. Convergence of $\Delta\Delta G$ with Simulation Time for Representative Mutations

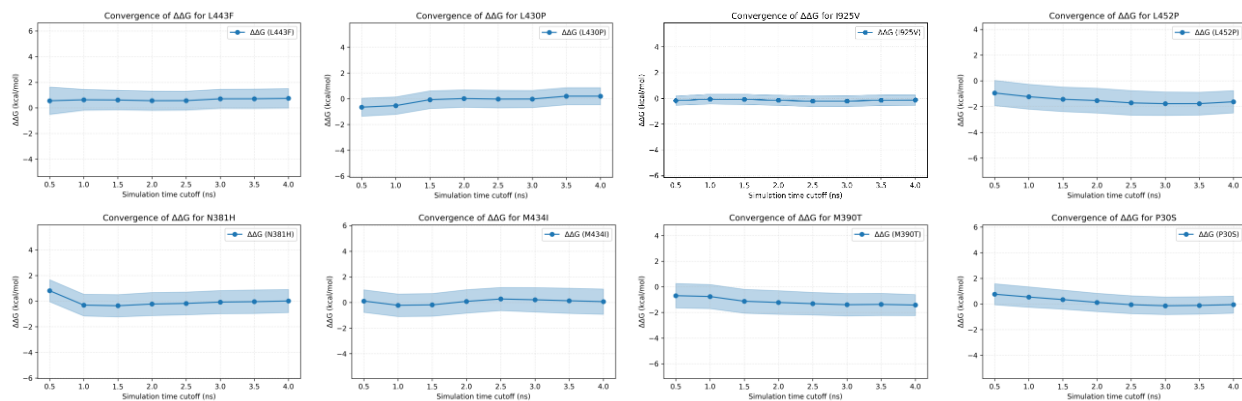
121



122



123



124

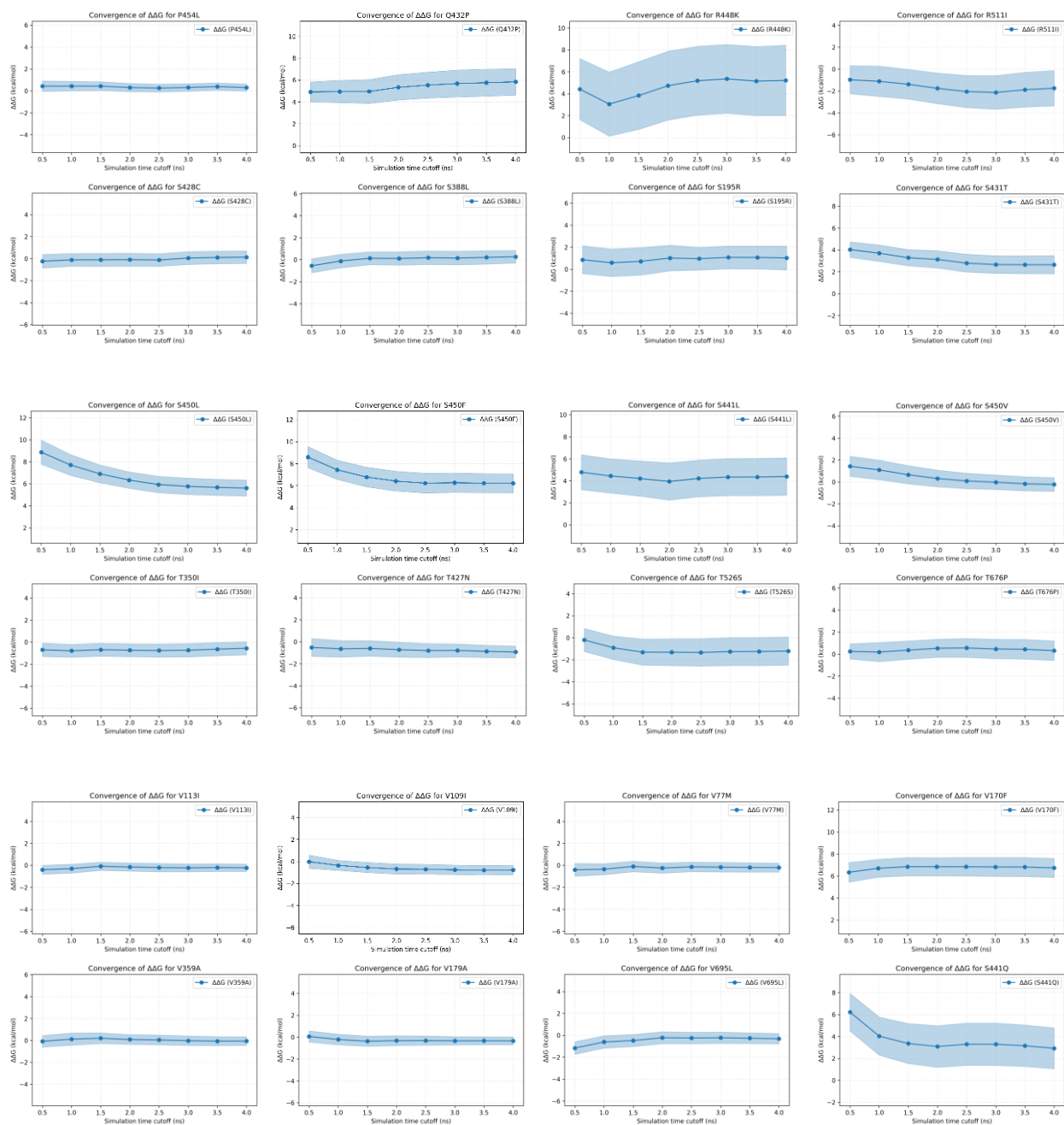


Figure S5.1 Convergence of $\Delta\Delta G$ for 48 mutations, where $\Delta\Delta G$ converged within 4 ns running. Each panel shows the running $\Delta\Delta G$ (blue circles) as a function of simulation time cutoff (0.5–4 ns) and the corresponding $\pm 1 \sigma$ uncertainty band (light-blue shaded region). All mutations reached $|\text{slope}| < 0.25 \text{ kcal/mol/ns}$ by 4 ns.

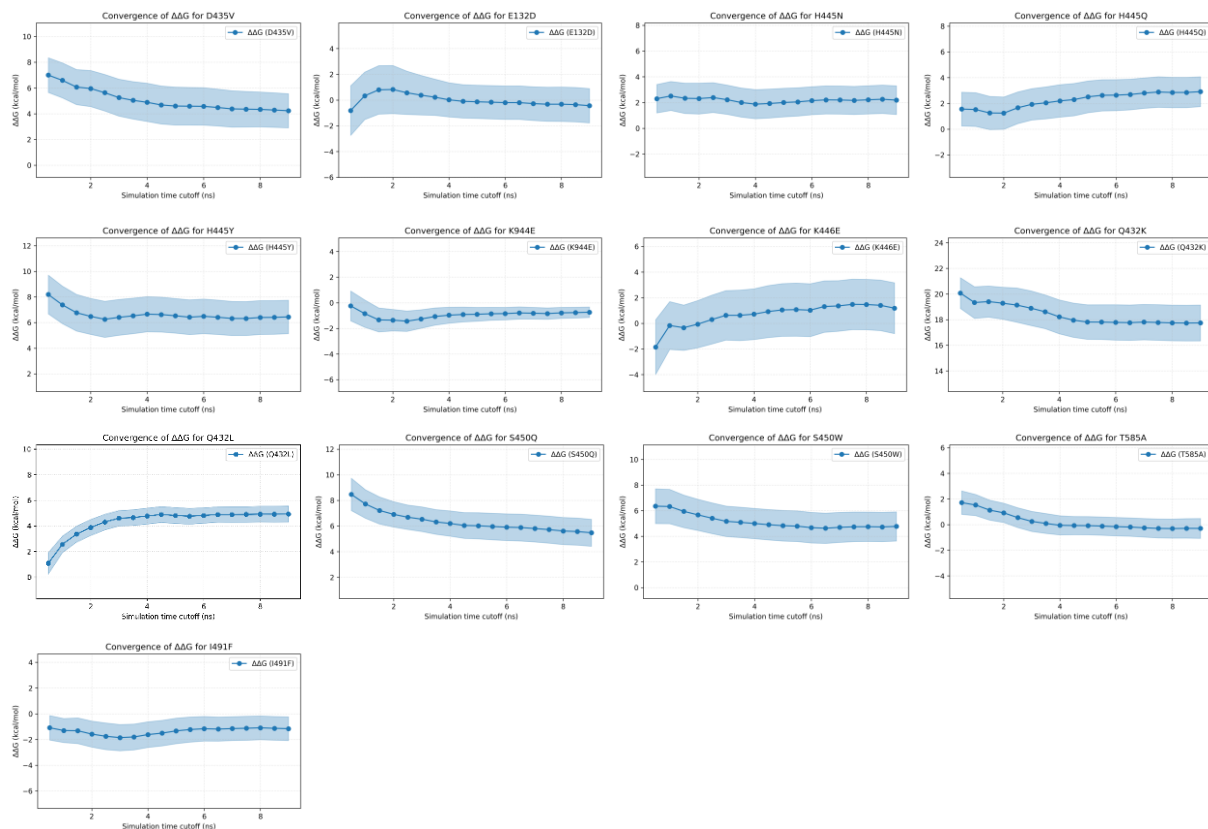


Figure S5. 2 Convergence of $\Delta\Delta G$ for 13 mutations, where $\Delta\Delta G$ converged within the extended 9 ns running.

Table S5. 1 $\Delta\Delta G$ Convergence slopes over the second half of simulation time for all mutations. (The unit for ddG-slope is kcal/mol/ns.)

Mutation	A334D	A451V	D270E	D435A	D435F	D435G	D435V
ddG-slope	-0.223	-0.042	-0.089	0.155	-0.229	-0.247	-0.095
Err	0.022	0.036	0.084	0.087	0.056	0.036	0.008
Mutation	E132D	E250G	G981D	G28R	G426S	H343Q	H445C
ddG-slope	-0.072	-0.069	0.045	0.036	0.124	-0.217	0.115
Err	0.006	0.075	0.021	0.003	0.040	0.049	0.078
Mutation	H445D	H445F	H445L	H445N	H445Q	H445R	H445Y
ddG-slope	-0.180	-0.103	0.173	0.050	0.097	0.111	-0.023
Err	0.080	0.043	0.026	0.014	0.011	0.079	0.018
Mutation	I925V	I491F	K944E	K446E	L430P	L443F	L452P
ddG-slope	0.071	0.041	0.033	0.087	0.179	0.100	0.054
Err	0.012	0.014	0.006	0.038	0.061	0.042	0.066
Mutation	M390T	M434I	N381H	P30S	P454L	Q432K	Q432L
ddG-slope	-0.068	-0.140	0.123	0.004	0.043	-0.021	0.038
Err	0.032	0.004	0.021	0.042	0.055	0.004	0.008
Mutation	Q432P	R448K	R511I	S195R	S388L	S428C	S431T

ddG-slope	0.194	0.023	0.237	0.049	0.066	0.168	-0.089
Err	0.025	0.101	0.097	0.058	0.035	0.051	0.046
Mutation	S441L	S441Q	S450F	S450L	S450Q	S450V	S450W
ddG-slope	0.098	-0.247	-0.020	-0.207	-0.138	-0.221	-0.002
Err	0.040	0.080	0.020	0.038	0.008	0.025	0.016
Mutation	T585A	T676P	T350I	T427N	T526S	V695L	V77M
ddG-slope	-0.060	-0.149	0.143	-0.096	0.068	-0.041	-0.042
Err	0.008	0.022	0.023	0.029	0.020	0.021	0.004
Mutation	V109I	V113I	V170F	V179A	V359A		
ddG-slope	-0.038	-0.005	-0.060	-0.016	-0.067		
Err	0.029	0.013	0.023	0.011	0.024		

135